Wave Speed

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Longer waves move faster at a fixed frequency.

Cases

Light waves need no medium.

Cases

Sound waves need a medium.

Checks

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- The angular frequency is $\omega = 2\pi f$.
- The wave number is $k = 2\pi/\lambda$.
- Speed is $v = \omega/k$.

Uncovered

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Where crests meet, the sum is larger.
Where a crest meets a trough, it cancels.

Summary

Frequency, wavelength and speed are linked by $v = \lambda f$.